purpose of the sampling strategy itself, the method of sampling by Hamiltonian trajectories seems less than robust.

Alternative families of methods exist which are based on extending the phase space of the system in such a way as to (a) disturb the conserved quantity $H = \mathrm{E}$, while (b) nonetheless preserving the phase space distribution. In general these methods involve introducing additional variables $\boldsymbol{\xi} = (\xi_1, \xi_2, \ldots, \xi_k)$ and additional equations of motion to drive them. The combined system is carefully designed to preserve an extended invariant distribution

$$\tilde{\rho}(\boldsymbol{q}, \boldsymbol{p}, \boldsymbol{\xi}) = \rho_\beta(\boldsymbol{q}, \boldsymbol{p}) \times \hat{\rho}(\boldsymbol{\xi}), \tag{8.1}$$

where the auxiliary density $\hat{\rho}$ has a simple form (typically, a multivariate Gaussian). Then it follows that

$$\int\int\int g(\boldsymbol{q}, \boldsymbol{p}) \tilde{\rho}(\boldsymbol{q}, \boldsymbol{p}, \boldsymbol{\xi}) \mathrm{d}\omega_{\boldsymbol{q}} \mathrm{d}\omega_{\boldsymbol{p}} \mathrm{d}^k \boldsymbol{\xi} \propto \int\int g(\boldsymbol{q}, \boldsymbol{p}) \rho_\beta(\boldsymbol{q}, \boldsymbol{p}) \mathrm{d}\omega_{\boldsymbol{q}} \mathrm{d}\omega_{\boldsymbol{p}}.$$

Assuming ergodicity of the coupled system it is then possible to calculate averages with respect to the Gibbs-Boltzmann measure by direct averaging along trajectories of the extended system.

The assumption of ergodicity is a major one (just as difficult to demonstrate as would be the corresponding assumption for a Hamiltonian system); indeed it is likely that it fails generically, and there are certainly known cases where thermodynamic averages computed using deterministic thermostats are incorrect. However, deterministic methods are nonetheless often used in molecular simulation, suggesting that, for the systems involved, averages can be recovered with sufficient accuracy starting from most points of phase space.

8.1.1 Nosé Dynamics: A Deterministic Thermostat

The simplest scheme of this type is the Nosé-Hoover method [182, 284–286]. There is just one auxiliary variable ξ and the extended system takes the form

$$\frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}t} = \boldsymbol{M}^{-1}\boldsymbol{p}, \tag{8.2}$$

$$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = -\nabla U(\boldsymbol{q}) - \xi \boldsymbol{p}, \tag{8.3}$$

$$\mu \frac{\mathrm{d}\xi}{\mathrm{d}t} = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p} - \mathrm{N_d} \beta^{-1}. \tag{8.4}$$

We have deliberately written $\mathrm{N_d}$. The number of degrees of freedom in the last equation must be used so that the equation is in balance at thermal equilibrium.